

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise AI platforms and multimodal systems. Specialized in PyTorch foundation model training, audio digital signal processing (DSP/ASR), and low-latency inference. Active contributor to Spring AI, bridging multimodal machine learning research with production-grade architectures.

Skills

Languages & Backend: Python, Java, C++, SQL, FastAPI, Spring Boot, PostgreSQL, REST APIs

AI & ML Engineering: Multimodal AI, Audio Signal Processing (DSP), Model Training & SFT, Computer Vision, Vector Embeddings

AI & ML Frameworks: PyTorch, Scikit-Learn, OpenCV, torchaudio, librosa, Hugging Face, vLLM, Triton

Platforms & Models: Whisper, IndicConformer, Claude, Vertex AI, Azure OpenAI, Qwen, TinyStories, Llama

Infrastructure & Security: Linux, Docker, Kubernetes, Kaggle Competitions, CI/CD, LLM-as-Judge, WebSockets

Work Experience

Bigscal Technologies Pvt. Ltd. (Client: TalenTeam Ltd., UK)

AI/ML Engineer

Oct 2024 - Present

Remote

- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
- Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by **70%**.
- Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned dense embeddings using PyTorch, writing custom CUDA kernels to optimize low-latency tensor operators.
- Developed the Prompt Lab evaluation control plane to collect multi-turn reasoning trajectories and human preference annotations, establishing a scalable Human Feedback (RLHF) Pipeline.
- Implemented a hierarchical multimodal document extraction pipeline indexing 10,000+ technical assets using GROBID, pdfplumber, and PGVector (90%+ Recall@10).
- Secured agentic workflows against OWASP LLM vulnerabilities using phantom-token context propagation, runtime tool monitoring, and trained token-classification models for automated PII redaction.

Healthray (Bigscal Technologies)

AI/ML Engineer Intern

Apr 2024 - Sep 2024

Surat, India

- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Mini Language Model (~85M Parameter Architecture) | Personal Research

Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.

Python, PyTorch, CUDA, Triton, Tesla P100/T4

AI Agent Security Red Teaming - OpenAI | Kaggle Researcher

Competed in the OpenAI red-teaming grand challenge on Kaggle, designing adversarial multi-modal evaluation pipelines and test cases across 2,000+ agent scenarios.

Python, PyTorch, Scikit-Learn, Kaggle

Modular Agent Harness & Memory System ("Brain") | Open Source

Architect of an open-source plug-and-play agentic framework designed with a formal ReAct loop (Thought-Action-Observation), tactical vector tool routing, and dynamic MCP settings caching.

Java, Spring Boot, PostgreSQL, WebSockets

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University

2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)

Microsoft

Microsoft Certified: Azure Fundamentals (AZ-900)

Microsoft